

ALEJANDRO MEDINA SANDÍN

Office contact information

Department of Economics
University of Verona
Via Cantarane 24
37129 Verona, Italy

Contact information

alejandro.medina@univr.it
alejandromedinasandin.com

Advisor

Paolo Pertile
University of Verona
paolo.pertile@univr.it

Placement Officers

Chiara Costi, chiara.costi@univr.it
Simone Quercia, simone.quercia@univr.it

EDUCATION

PhD in Economics and Finance, University of Verona	2027 (expected)
– Visiting Scholar, University of Mannheim	2023–2024
MSc in Economics, Universitat de Barcelona	2022
BSc in Economics, Universitat Pompeu Fabra	2020

REFERENCES

Paolo Pertile
University of Verona
paolo.pertile@univr.it

Albert Banal-Estañol
Universitat Pompeu Fabra
albert.banalestanol@upf.edu

Michelle Sovinsky
University of Mannheim
michelle.sovinsky@uni-mannheim.de

FIELDS

Primary Fields: Empirical Industrial Organization, Competition Policy and Antitrust

Secondary Fields: Health Economics, Applied Econometrics, AI Applied to Economics

PERSONAL DETAILS

Citizenship: Spanish

Languages: Spanish (native), Catalan (native), English (C1), Italian (B2)

TEACHING EXPERIENCE

Artificial Intelligence for Research (main instructor; 16-hour course for MSc and PhD students), University of Huelva	2026
Industrial Organization (Teaching Assistant), University of Verona	2023

RELEVANT POSITIONS

Intern, Berkeley Research Group, Brussels (European Competition practice)	2024–2025
---	-----------

FELLOWSHIPS AND AWARDS

- Innova Award to the Young Researcher with the Best R&D&i Project 2026
(PhD), Universitat Rovira i Virgili
Awarded for the paper “Pay for Delay: Preventing Competition in Pharmaceutical Markets”, within the PhD-Student Workshop in Industrial and Public Economics (WIPE).
- Young Talent Competition Award, Lear Competition Festival 2025
Awarded for the paper “Pay for Delay: Preventing Competition in Pharmaceutical Markets”; prize for young researchers in competition law, economics, and regulation.
- Doctoral Fellowship, University of Verona 2022–2026

ADDITIONAL TRAINING

- Demand Estimation (A. Pakes, J. Gortmaker), Causal Inference Workshop 2024
- CRESSE Summer School on Competition Policy and Regulation 2023
- Center for Competition Policy Summer School in Platform Economics 2023

PRESENTATIONS

2026: XL Jornadas de Economía Industrial (Málaga), EARIE Annual Conference (Mannheim), BSE PhD Jamboree, Barcelona School of Economics (Barcelona), EuHEA Seminar Series (online), CRES Seminar, Universitat Pompeu Fabra (Barcelona), WIPE Workshop on Industrial and Public Economics (Reus)

2025: XXXIX Jornadas de Economía Industrial (Santander), Lear Competition Festival (Rome), EARIE Annual Conference (Valencia), Centre for Competition Policy Seminar (London), CCP Conference on Frontiers of Competition and Regulation (London)

WORKING PAPERS

Pay for Delay: Preventing Competition in Pharmaceutical Markets (Job Market Paper)
Revise and Resubmit, Journal of Industrial Economics

Pay-for-Delay agreements occur when a branded pharmaceutical firm compensates a generic competitor to delay market entry, raising concerns about competition and consumer welfare. This paper quantifies the consumer harm from the agreement that delayed generic entry into the U.S. Testosterone Replacement Therapy market from 2006 to 2015. I estimate a two-level nested logit demand model on monthly product-level data, recover marginal costs from Bertrand-Nash first-order conditions, and simulate a counterfactual in which generics enter before 2015 by adding fictitious generic products to the choice set and solving for new equilibrium prices that account for the responses of all incumbent firms. Earlier generic entry would have made alternatives available at prices 5–11% below branded levels. Total consumer damages amount to \$326 million over the delay period, an 18% increase in consumer surplus. The supply-side analysis confirms that the agreement was individually rational for all parties, sustained by the direct transfer of monopoly rents rather than by strategic advantages from delayed entry.

AI Meets Econometrics: A Multi-Model Evaluation of LLM Pedagogical Quality

Revise and Resubmit, International Review of Economics Education

This paper evaluates the pedagogical quality of three frontier large language models – Claude Sonnet 4.5, GPT-5, and DeepSeek-V3 – as study aids for undergraduate econometrics. Twenty-nine questions from a standard undergraduate textbook, covering OLS, panel data, instrumental variables,

and time series, are submitted to each model under identical prompts. Responses are scored with a seven-dimension rubric that separates technical correctness from pedagogical quality. All three models score highly on accuracy, completeness, and terminology, so factual reliability no longer discriminates between frontier systems. The differences emerge in technical clarity, explanatory clarity, and pedagogical usefulness: Claude leads overall, especially on computational questions; GPT performs best on conceptual questions; and DeepSeek consistently underperforms despite comparable accuracy. The pedagogical risk of AI-assisted learning therefore lies not in incorrect content but in the quality of the explanation that accompanies correct answers, and the best model depends on the type of task.

When Does the “Generic Paradox” Occur? Evidence from Therapeutic Class Heterogeneity

Standard theory predicts that prices fall as the number of competitors increases, yet the evidence on how generic entry affects branded drug prices is mixed: some studies find that branded prices rise after entry (the “Generic Paradox”), others that they fall. This paper asks whether heterogeneity across therapeutic classes explains these divergent findings and offers the broadest empirical test of the paradox to date. Using data on more than 475 branded products in two major therapeutic classes (C09 and C10), I apply the Callaway and Sant’Anna (2021) difference-in-differences estimator to measure the effect of generic entry on branded prices and market shares. The results diverge sharply: in C10 branded prices rise by about 50% after 30 quarters, while in C09 they fall by 20–40% after 20 quarters. Branded market share erodes in both classes, but within 10 quarters in C09 versus about 20 in C10. The Generic Paradox is therefore therapeutic-class specific, and the mixed prior evidence reflects genuine heterogeneity across markets rather than conflicting estimates of a single effect.

AI Meets Antitrust: How Large Language Models Reshape Market Concentration

Large language models are becoming a primary interface for consumer product discovery, yet little is known about how their recommendations relate to actual market structure. This paper provides the first empirical benchmark of LLM brand recommendations against market shares. Using 1,429 responses from three frontier models (GPT-4o, Claude Sonnet 4, Gemini 2.5 Flash Lite) across sixteen U.S. consumer categories, I compare the implied concentration of recommendations with retail-value shares from Euromonitor Passport 2024. In fragmented markets such as skincare, LLMs concentrate recommendations on fewer brands than market shares would suggest; in concentrated markets such as headphones, they spread them across more brands. Both patterns reflect a single phenomenon: LLM recommendations live in a concentration band roughly one third as wide as the real market’s, regardless of where the market sits. Top recommendations also tilt toward specialist and premium brands that hold only two thirds of the share of actual category leaders. Findings are stable across the three models.

Can Divestitures Undo Cartel Damage? Evidence from the Teva-Allergan Merger

This paper examines whether structural remedies can reverse the price effects of collusion. It exploits a unique overlap in the U.S. generic pharmaceutical industry, where a subset of molecules were simultaneously affected by the 2013–2015 generic price-fixing cartel and subject to the mandatory divestitures imposed by the Federal Trade Commission in the 2016 Teva-Allergan merger. By comparing the price evolution of molecules that were cartelized only, divested only, or both, the paper evaluates whether the divestitures restored competitive pricing precisely in the markets where prior collusion had elevated prices above competitive levels.

OTHER INFORMATION

Software: Stata, MATLAB, Python, R, L^AT_EX